Unit Vectors

A *unit vector* is a vector whose magnitude is 1. The vectors $\vec{i}$, $\vec{j}$, and $\vec{k}$ are unit vectors in the directions of the coordinate axes. It is often helpful to find a unit vector in the same direction as a given vector $\vec{v}$. Suppose that $\|\vec{v}\| = 10$; a unit vector in the same direction as $\vec{v}$ is $\vec{v}/10$. In general, a unit vector in the direction of any nonzero vector $\vec{v}$ is

$$\vec{u} = \frac{\vec{v}}{\|\vec{v}\|}.$$

Example 5 Find a unit vector, $\vec{u}$, in the direction of the vector $\vec{v} = \vec{i} + 3\vec{j}$.

Solution If $\vec{v} = \vec{i} + 3\vec{j}$, then $\|\vec{v}\| = \sqrt{1^2 + 3^2} = \sqrt{10}$. Thus, a unit vector in the same direction is given by

$$\vec{u} = \frac{\vec{v}}{\sqrt{10}} = \frac{1}{\sqrt{10}}(\vec{i} + 3\vec{j}) = \frac{1}{\sqrt{10}}\vec{i} + \frac{3}{\sqrt{10}}\vec{j} \approx 0.32\vec{i} + 0.95\vec{j}.$$

Example 6 Find a unit vector at the point (x, y, z) that points radially outward away from the origin.

Solution The vector from the origin to (x, y, z) is the position vector

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}.$$

Thus, if we put its tail at (x, y, z) it will point away from the origin. Its magnitude is

$$\|\vec{r}\| = \sqrt{x^2 + y^2 + z^2},$$

so a unit vector pointing in the same direction is

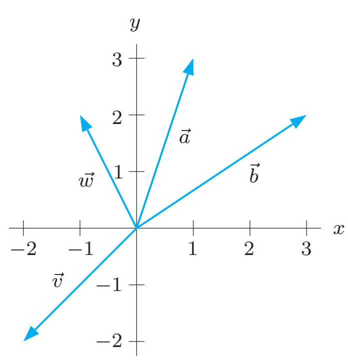
$$\frac{\vec{r}}{\|\vec{r}\|} = \frac{x\vec{i} + y\vec{j} + z\vec{k}}{\sqrt{x^2 + y^2 + z^2}} = \frac{x}{\sqrt{x^2 + y^2 + z^2}}\vec{i} + \frac{y}{\sqrt{x^2 + y^2 + z^2}}\vec{j} + \frac{z}{\sqrt{x^2 + y^2 + z^2}}\vec{k}.$$

Exercises and Problems for Section 13.1

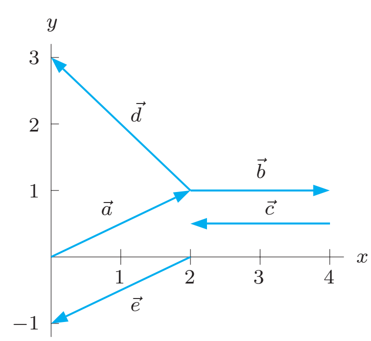
Exercises

In Exercises 1–6, resolve the vectors into components.

1.



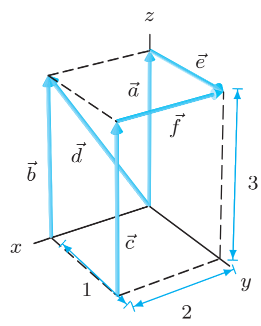
2.



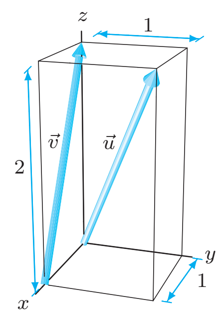
3. A vector starting at the point $Q = (4, 6)$ and ending at the point $P = (1, 2)$.

4. A vector starting at the point $P = (1, 2)$ and ending at the point $Q = (4, 6)$.

5.



6.



For Exercises 7–14, perform the indicated computation.

7. $(4\vec i + 2\vec j) - (3\vec i - \vec j)$

8. $(\vec i + 2\vec j) + (-3)(2\vec i + \vec j)$

9. $-4(\vec i - 2\vec j) - 0.5(\vec i - \vec k)$

10. $2(0.45\vec i - 0.9\vec j - 0.01\vec k) - 0.5(1.2\vec i - 0.1\vec k)$

11. $(3\vec i - 4\vec j + 2\vec k) - (6\vec i + 8\vec j - \vec k)$

12. $(4\vec i - 3\vec j + 7\vec k) - 2(5\vec i + \vec j - 2\vec k)$

13. $(0.6\vec i + 0.2\vec j - \vec k) + (0.3\vec i + 0.3\vec k)$

14. $\frac{1}{2}(2\vec i - \vec j + 3\vec k) + 3(\vec i - \frac{1}{6}\vec j + \frac{1}{2}\vec k)$

In Exercises 15–19, find the length of the vectors.

15. $\vec v = \vec i - \vec j + 2\vec k$

16. $\vec z = \vec i - 3\vec j - \vec k$

17. $\vec v = \vec i - \vec j + 3\vec k$

18. $\vec v = 7.2\vec i - 1.5\vec j + 2.1\vec k$

19. $\vec v = 1.2\vec i - 3.6\vec j + 4.1\vec k$

For Exercises 20–25, perform the indicated operations on the following vectors:

$$\vec a = 2\vec j + \vec k, \quad \vec b = -3\vec i + 5\vec j + 4\vec k, \quad \vec c = \vec i + 6\vec j,$$

$$\vec x = -2\vec i + 9\vec j, \quad \vec y = 4\vec i - 7\vec j, \quad \vec z = \vec i - 3\vec j - \vec k.$$

20. $4\vec z$

21. $5\vec a + 2\vec b$

22. $\vec a + \vec z$

23. $2\vec c + \vec x$

24. $2\vec a + 7\vec b - 5\vec z$

25. $\|\vec y - \vec x\|$

26. **(a)** Draw the position vector for $\vec v = 5\vec i - 7\vec j$.
(b) What is $\|\vec v\|$?
(c) Find the angle between $\vec v$ and the positive x-axis.

27. Find the unit vector in the direction of $0.06\vec i - 0.08\vec k$.

28. Find the unit vector in the opposite direction to $\vec i - \vec j + \vec k$.

29. Find a unit vector in the opposite direction to $2\vec i - \vec j - \sqrt{11}\vec k$.

30. Find a vector with length 2 that points in the same direction as $\vec i - \vec j + 2\vec k$.

Problems

31. Find the value(s) of a making $\vec v = 5a\vec i - 3\vec j$ parallel to $\vec w = a^2\vec i + 6\vec j$.

32. **(a)** Find a unit vector from the point $P = (1, 2)$ and toward the point $Q = (4, 6)$.
(b) Find a vector of length 10 pointing in the same direction.

33. If north is the direction of the positive y-axis and east is the direction of the positive x-axis, give the unit vector pointing northwest.

34. Resolve the following vectors into components:

(a) The vector in 2-space of length 2 pointing up and to the right at an angle of $\pi/4$ with the x-axis.
(b) The vector in 3-space of length 1 lying in the xz-plane pointing upward at an angle of $\pi/6$ with the positive x-axis.

35. **(a)** From Figure 13.16, read off the coordinates of the five points, A, B, C, D, E, and thus resolve into components the following two vectors: $\vec u = (2.5)\overrightarrow{AB} + (-0.8)\overrightarrow{CD}$, $\vec v = (2.5)\overrightarrow{BA} - (-0.8)\overrightarrow{CD}$
(b) What is the relation between $\vec u$ and $\vec v$? Why was this to be expected?

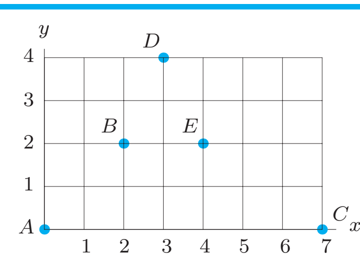


Figure 13.16

36. Find the components of a vector $\vec p$ that has the same direction as $\overrightarrow{EA}$ in Figure 13.16 and whose length equals two units.

37. For each of the four statements below, answer the following questions: Does the statement make sense? If yes, is it true for all possible choices of $\vec a$ and $\vec b$? If no, why not?

(a) $\vec a + \vec b = \vec b + \vec a$
(b) $\vec a + \|\vec b\| = \|\vec a + \vec b\|$
(c) $\|\vec b + \vec a\| = \|\vec a + \vec b\|$
(d) $\|\vec a + \vec b\| = \|\vec a\| + \|\vec b\|$.

38. Two adjacent sides of a regular hexagon are given as the vectors $\vec{u}$ and $\vec{v}$ in Figure 13.17. Label the remaining sides in terms of $\vec{u}$ and $\vec{v}$.

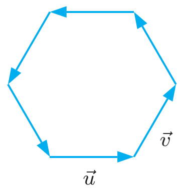


Figure 13.17

39. For what values of t are the following pairs of vectors parallel?

(a) $2\vec{i} + (t^2 + \frac{2}{3}t + 1)\vec{j} + t\vec{k}$, $\quad 6\vec{i} + 8\vec{j} + 3\vec{k}$

(b) $t\vec{i} + \vec{j} + (t-1)\vec{k}$, $\quad 2\vec{i} - 4\vec{j} + \vec{k}$

(c) $2t\vec{i} + t\vec{j} + t\vec{k}$, $\quad 6\vec{i} + 3\vec{j} + 3\vec{k}$.

40. Find all vectors $\vec{v}$ in 2 dimensions having $\|\vec{v}\| = 5$ such that the $\vec{i}$-component of $\vec{v}$ is $3\vec{i}$.

41. Find all vectors $\vec{v}$ in the plane such that $\|\vec{v}\| = 1$ and $\|\vec{v} + \vec{i}\| = 1$.

42. Figure 13.18 shows a molecule with four atoms at O, A, B and C. Check that every atom in the molecule is 2 units away from every other atom.

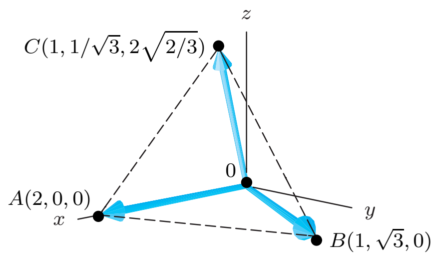


Figure 13.18

43. Show that the medians of a triangle intersect at a point $\frac{1}{3}$ of the way along each median from the side it bisects.

Strengthen Your Understanding

In Problems 44–47, explain what is wrong with the statement.

44. If $\|\vec{u}\| = 1$ and $\|\vec{v}\| > 0$, then $\|\vec{u} + \vec{v}\| \geq 1$.

45. The vector $c\vec{u}$ has the same direction as $\vec{u}$.

46. $\|\vec{v} - \vec{u}\|$ is the length of the shorter of the two diagonals of the parallelogram determined by $\vec{u}$ and $\vec{v}$.

47. Given three vectors $\vec{u}$, $\vec{v}$, and $\vec{w}$, if $\vec{u} + \vec{w} = \vec{u}$ then it is possible for $\vec{v} + \vec{w} \neq \vec{v}$.

In Problems 48–50, give an example of:

48. A vector $\vec{v}$ of length 2 with a positive $\vec{k}$-component and lying on a plane parallel to the yz-plane.

49. Two unit vectors $\vec{u}$ and $\vec{v}$ for which $\vec{v} - \vec{u}$ is also a unit vector.

50. Two vectors $\vec{u}$ and $\vec{v}$ that have difference vector $\vec{w} = 2\vec{i} + 3\vec{j}$.

Are the statements in Problems 51–60 true or false? Give reasons for your answer.

51. There is exactly one unit vector parallel to a given nonzero vector $\vec{v}$.

52. The vector $\frac{1}{\sqrt{3}}\vec{i} + \frac{-1}{\sqrt{3}}\vec{j} + \frac{2}{\sqrt{3}}\vec{k}$ is a unit vector.

53. The length of the vector $2\vec{v}$ is twice the length of the vector $\vec{v}$.

54. If $\vec{v}$ and $\vec{w}$ are any two vectors, then $\|\vec{v} + \vec{w}\| = \|\vec{v}\| + \|\vec{w}\|$.

55. If $\vec{v}$ and $\vec{w}$ are any two vectors, then $\|\vec{v} - \vec{w}\| = \|\vec{v}\| - \|\vec{w}\|$.

56. The vectors $2\vec{i} - \vec{j} + \vec{k}$ and $\vec{i} - 2\vec{j} + \vec{k}$ are parallel.

57. The vector $\vec{u} + \vec{v}$ is always larger in magnitude than both $\vec{u}$ and $\vec{v}$.

58. For any scalar c and vector $\vec{v}$ we have $\|c\vec{v}\| = c\|\vec{v}\|$.

59. The displacement vector from $(1,1,1)$ to $(1,2,3)$ is $-\vec{j} - 2\vec{k}$.

60. The displacement vector from (a,b) to (c,d) is the same as the displacement vector from (c,d) to (a,b).

13.2 VECTORS IN GENERAL

Besides displacement, there are many quantities that have both magnitude and direction and are added and multiplied by scalars in the same way as displacements. Any such quantity is called a *vector* and is represented by an arrow in the same manner we represent displacements. The length of the arrow is the *magnitude* of the vector, and the direction of the arrow is the direction of the vector.

Velocity Versus Speed

The speed of a moving body tells us how fast it is moving, say 80 km/hr. The speed is just a number; it is therefore a scalar. The velocity, on the other hand, tells us both how fast the body is moving and the direction of motion; it is a vector. For instance, if a car is heading northeast at 80 km/hr, then its velocity is a vector of length 80 pointing northeast.